

PromptoPal

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Problem Statement and Abstract

Over the last two decades, autism has been identified with striking and rising frequency, especially in places like the United States of America. A common problem with most kids on the lower end of the spectrum is task initiation. Abhinav has first hand experience with people like this because his older brother is on the autism spectrum and experiences difficulty with task initiation. He does not begin activities (showering, puzzles, swimming, cycling) unless explicitly prompted. Family members cannot be present 24/7. Existing solutions (timers, phone alerts, human caregivers) are either ignored, inaccessible, or unsustainable.

Leaving people diagnosed with autism to the care of a nurse isn't always the most safe or practical due to the uncertainty of what they can do and the high cost (often goes up to more than \$80k per year).

Our Solution

We are in the process of building an autonomous mobile robot called PromptoPal that will search Abhinav's house at scheduled times, locate his brother, facially and verbally confirm his identity, and deliver task prompts with supportive music.

This would potentially be a lot more effective in comparison to a phone alert or timer that most people use in their day to day lives because autistic kids

Subsystem	Components	Purpose
Brain	Raspberry Pi 5 (4GB) + 32 GB MicroSD card	Runs Python-based control logic, computer vision, and speech
Vision	Pi Camera Module 3 + 300 mm ribbon cable	Person detection via OpenCV; ArUco marker localization
Audio Input	USB Conference microphone	Captures voice response to "What's your name?"
Audio Output	PAM8403 amplifier + 4Ω 3W speaker + 3.5mm cable	Delivers verbal prompts and plays calming music
Navigation	HC-SR04 Ultrasonic sensors for obstacle avoidance	Prevents collisions during movement
Power	12V 10Ah Lithium LiFePO4 Deep cycle rechargeable battery + Buck converter (12V→5.1V)	Mobile power for all electronics
Mobility	Arduino Nano Microcontroller, 12V DC 200RPM Turbo Worm Geared Motor, DC Motor Driver BTS7960	Physical movement through house

Software Stack:

OS: Raspberry Pi OS (Linux)

Languages: Python 3

Python over C++: Faster iteration and a lot more mature libraries on Pi

Libraries: OpenCV (Vision), pytsx3 (Speech), SpeechRecognition (voice), RPi.GPIO (motor control)

Challenge	Implementation
Indoor navigation without encoders	Pre-taught timed paths + ArUco markers for final positioning correction
Obstacle avoidance	Ultrasonic sensor triggers stop/recovery routine
Person detection	OpenCV Haar cascade classifier identifies

	humans in frame
User identification	Robot asks "What's your name?"; keyword spotting matches brother's name
Speech in noisy environments	Omnidirectional USB microphone with basic noise filtering
Power stability	LM2596 buck converter with input capacitors filters motor noise
Multi-room search	Sequentially visits pre-taught locations until person found or list exhausted

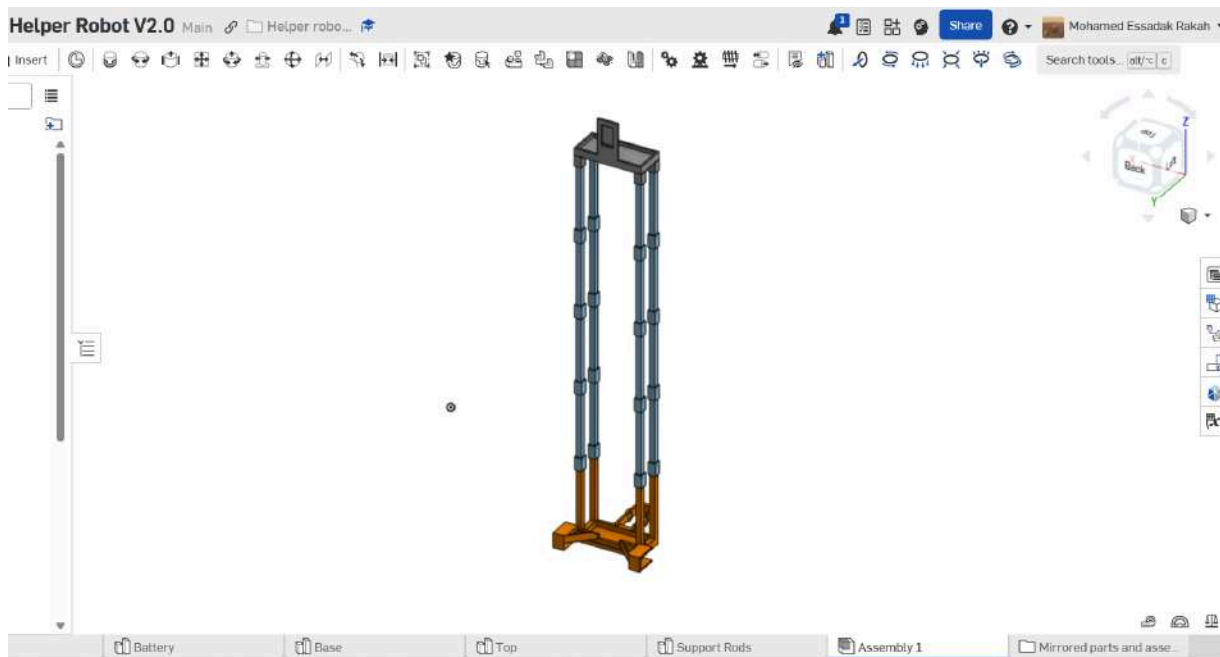
Limitations	Why it matters
Rule-based behavior	Cannot adapt if brother's routine or responses change
Fixed navigation paths	Only works in one house; furniture moves break routes
No learning from interaction	Each day is the same as the last—no improvement
Single-user focus	Cannot generalize to other individuals
Voice recognition brittle	Fails in high noise or if brother is non-verbal at that moment

Programming Architecture:

Scheduler	Python schedule library triggers room-by-room searches at preset times
Navigation	Teach-and-replay: manually driven paths recorded as timed motor commands; ArUco markers provide visual correction at each room
Vision	OpenCV Haar cascades for person detection; cv2.aruco for marker localization
Speech	pyttsx3 for text-to-speech; lightweight keyword spotting for name recognition

	(offline)
Obstacle Avoidance	HC-SR04 ultrasonic sensor runs parallel loop; interrupts motor control if distance < 20cm
State Machine	Tracks: SEARCH → DETECT → ASK → CONFIRM → PROMPT → RETURN
Error Handling	If marker not found: 10s spin search → return to charger; if no person: proceed to next room

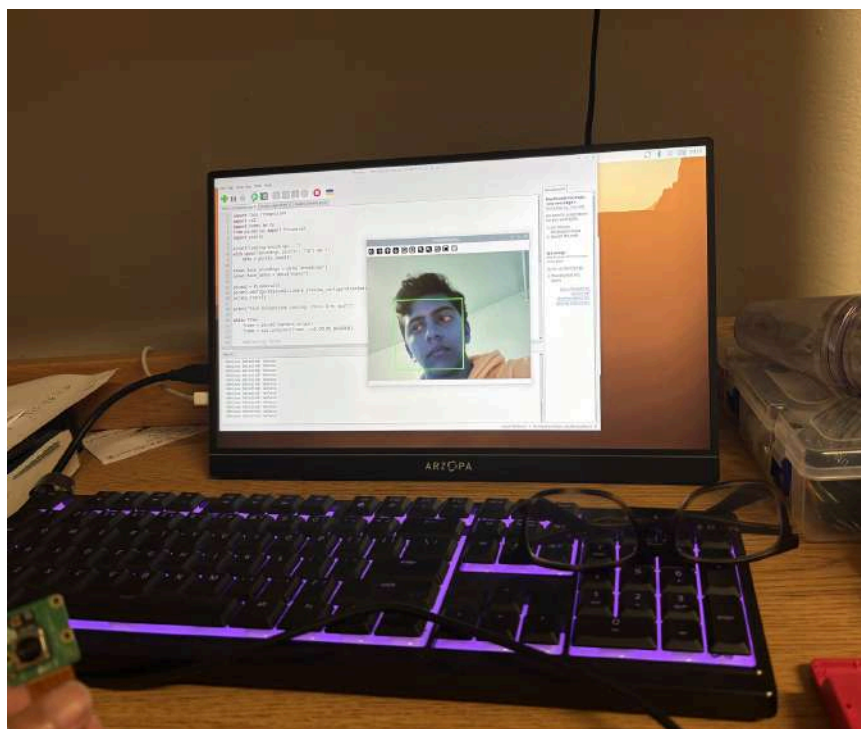
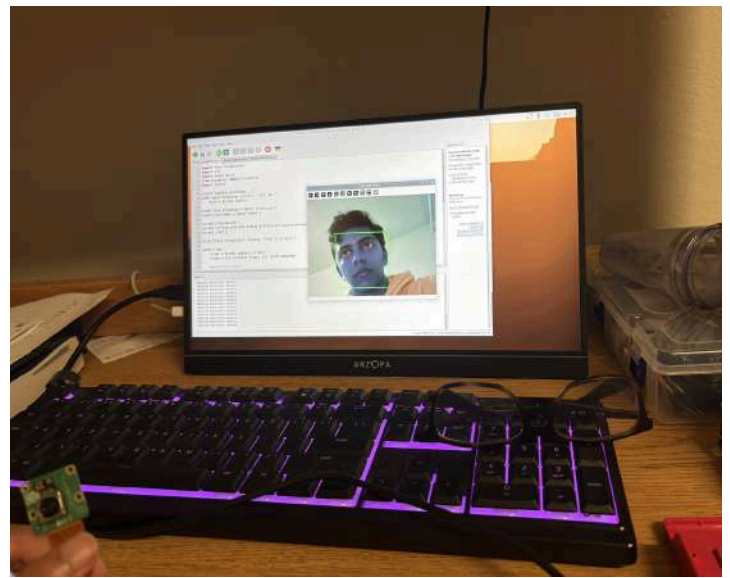
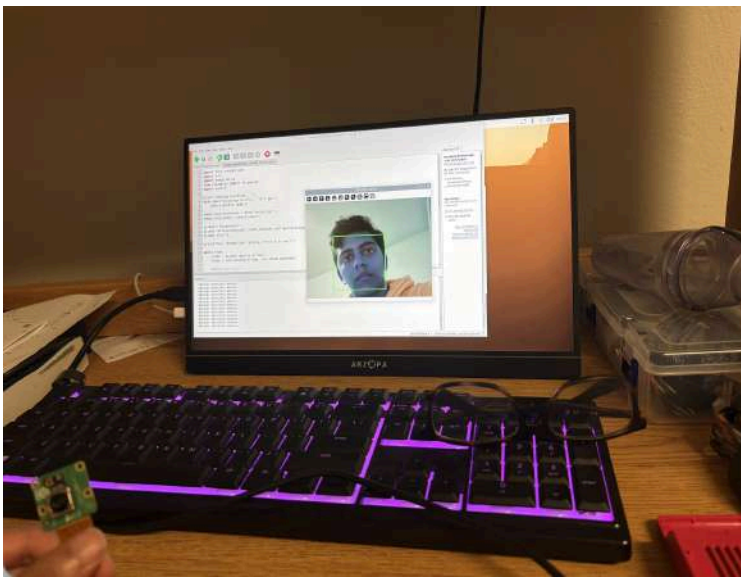
OnShape Software to create the 3D model



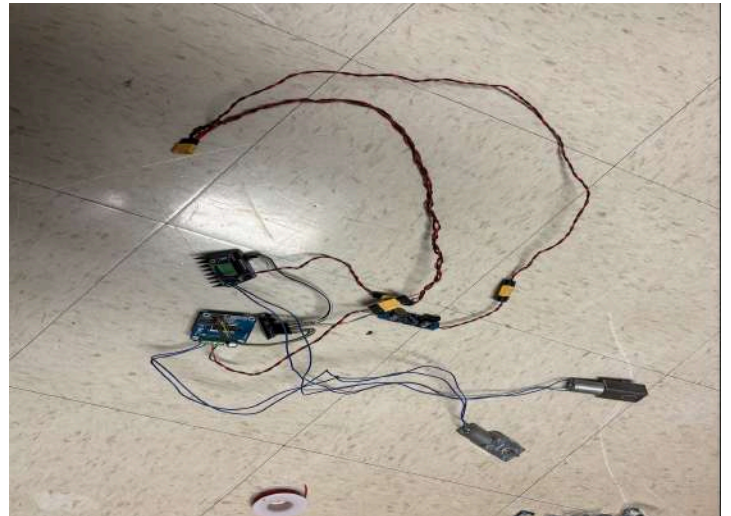
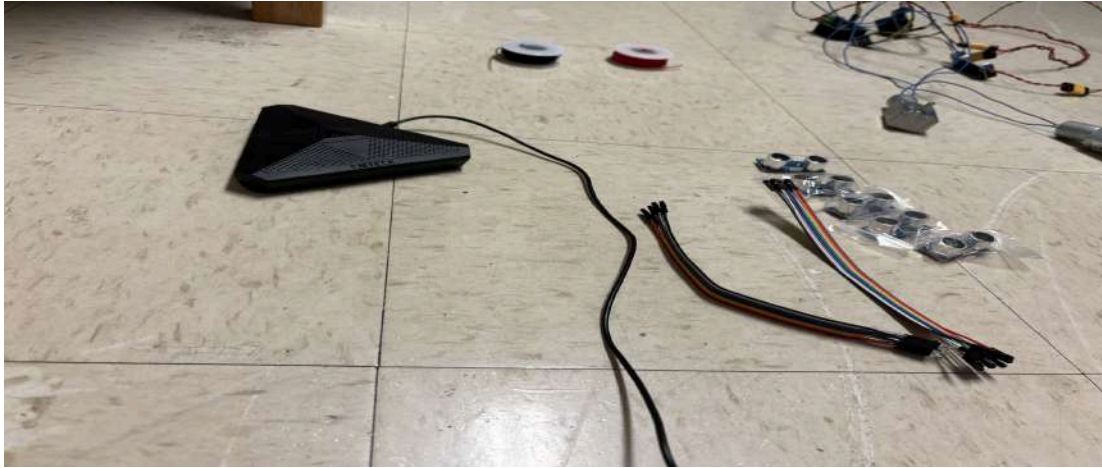
3D printed base of the robot, support rods to be attached to increase the height up to 1 meter tall

Facial recognition to identify the person:

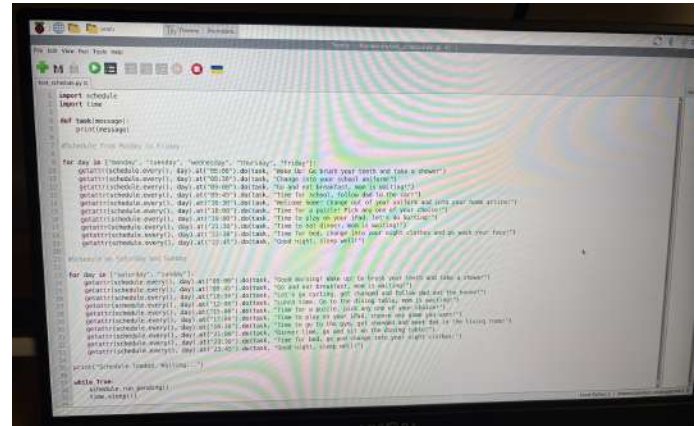
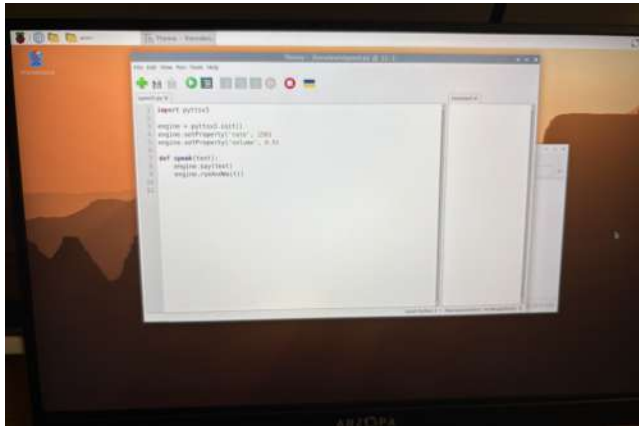
Composed of 3 files, `image_capture.py` which allows photos to be clicked, `model_training.py` which trains the Pi to learn and manipulate every pixel in the images taken to create a solid reference, `face_recognition.py` which operates in real time and a green square is drawn around the face when identified to be the same as the ones trained in the images captured.



Other equipment like motors, speakers, ultrasonic sensors, LiFePO4 battery



Code written to build the schedule:



Drawbacks of Prototypes

Right now, the robot is very hard coded and there is no AI integrated into it. It will work only for a single face in a single environment. Only face_recognition.py will run on the when it is on the robot, image_capture.py will not, which means the robot isn't actively taking pictures as it is moving. It is using images that are already captured as a reference when searching, but if a new face is to be detected instead, it will not function as intended. The model is also very skinny and could collapse with the slightest of interactions. It doesn't look humanoid, but as a total machine, which might not work entirely with autistic kids as they only listen to those who they trust. It also does not learn over time which is something that is needed for autistic kids. There are various scenarios in which they might not respond instantly to a command from the robot, or they might refuse to abide by their routine despite being reminded by the robot. In all these scenarios, the robot is expected to maintain patience and stay within the vicinity of the person to ensure they are alright. This is something our robot lacks at the moment, but we are working on to improve.

Future Goals:

We want to build an app in the future that gets linked to the robot via bluetooth. Through this app, the user can enter all the details required such as their schedule, their face, their voice, house layout, etc. This app will build frameworks for the robot to follow and ensure that it works for anyone in any environment. AI integration will also be implemented, and several cloud APIs will be used to ensure that various functions like speech, audio, etc are in the robot.